

EDUGENIE PROJECT – TEAM DETAILS

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EDUGENIE – GOOGLE GEMINI POWERED LEARNING ASSISTANT

PHASE 1: BRAINSTORMING & IDEATION

1. Introduction

Brainstorming and ideation is the first phase of the EduGenie project. This phase focuses on identifying a real problem faced by students and finding a suitable technology-based solution. The team discussed different educational problems and explored how Artificial Intelligence could be used to make learning easier, faster, and more interactive. The main objective was to develop an idea that would be useful for students and practical to implement as a project.

2. Need for the Project

Students often face difficulties when they study independently. They may not understand difficult concepts, may need simple explanations, or may have questions outside classroom hours. Searching through many websites can also take time and may provide information that is too complicated for the student's level. Therefore, there is a need for a simple learning assistant that can provide quick and understandable educational support.

The growth of Artificial Intelligence provides an opportunity to address this problem. An AI-powered learning assistant can understand natural-language questions and generate responses based on the user's input. This idea became the foundation for the EduGenie project.

3. Problem Identification

During the discussion, the team identified several common learning problems. Students may hesitate to ask questions repeatedly, may have difficulty finding simple explanations, and may spend considerable time searching for suitable study materials. Students also have different learning speeds and may need explanations in a more understandable form.

The problem identified for the project was therefore: **Students need an easily accessible and simple AI-based learning assistant that can provide quick explanations and learning support.**

4. Brainstorming Process

The team conducted brainstorming sessions to generate possible solutions. Different ideas were discussed, including educational websites, question-answer systems, study assistants, revision tools, and AI-based learning applications. Each idea was considered based on usefulness, feasibility, technology requirements, and suitability for students.

The team then compared the ideas and focused on the possibility of using a Generative AI model to create a flexible learning assistant. Google Gemini was considered suitable because it can process natural-language questions and generate responses that can be used for educational assistance.

5. Initial Ideas Considered

- An online student question-and-answer platform.
- A digital study assistant for explaining difficult topics.

- An AI chatbot that could answer educational questions.
- A revision assistant that could provide summaries and explanations.
- A learning assistant powered by a modern Generative AI model.

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PHASE 1: BRAINSTORMING & IDEATION – CONTINUED

6. Selection of the EduGenie Idea

After comparing the possible ideas, the team selected an AI-powered learning assistant named **EduGenie**. The name represents an educational assistant that helps students learn with the support of Artificial Intelligence. The selected idea combines a simple student interface with the capabilities of Google Gemini.

EduGenie is planned to allow students to enter questions in natural language and receive AI-generated educational responses. The project aims to make learning support more accessible and reduce the difficulty of finding simple explanations.

7. Project Vision

The vision of EduGenie is to create a student-friendly learning assistant that can support learners whenever they need help understanding a topic. The system should provide information in a clear and understandable manner while keeping the interaction simple. The project is intended to demonstrate how Generative AI can be applied to an educational use case.

8. Target Users

The primary target users are students and learners who need additional support during self-study. The system can be useful for students who want quick explanations, clarification of concepts, or basic learning assistance. Teachers and project evaluators can also use the system demonstration to understand how AI can support education.

9. Main Features Identified During Ideation

- Simple and student-friendly user interface.
- Natural-language question input.
- AI-powered response generation using Google Gemini.
- Easy-to-understand explanations.
- Quick response to student questions.
- Support for learning and revision.
- Clear presentation of generated answers.
- Scope for adding more educational features in the future.

10. Expected Benefits

EduGenie is expected to provide students with quick access to learning assistance. It can reduce the time spent searching for basic explanations and can make difficult topics easier to approach. The project also demonstrates a practical application of Generative AI in education and gives students an opportunity to interact with modern AI technology.

11. Initial Project Scope

The initial scope of the project is to develop and demonstrate the core learning assistant workflow. The system will accept a student's question, process it through the AI-powered system, and display the generated response. Advanced features can be considered later depending on project requirements and available resources.

12. Conclusion

The brainstorming and ideation phase established the foundation of the EduGenie project. A real student learning problem was identified, several possible solutions were discussed, and an AI-powered learning assistant was selected as the final idea. The target users, project vision, major features, expected benefits, and initial scope were also identified. This finalized idea provides a clear starting point for the next phase, Requirement Analysis.